Platform frame

This is the most commonly used method in the UK. Each storey is framed with floor-to-ceiling height stud walls and the floor deck of one floor becomes the erection platform of the next, simplifying construction.

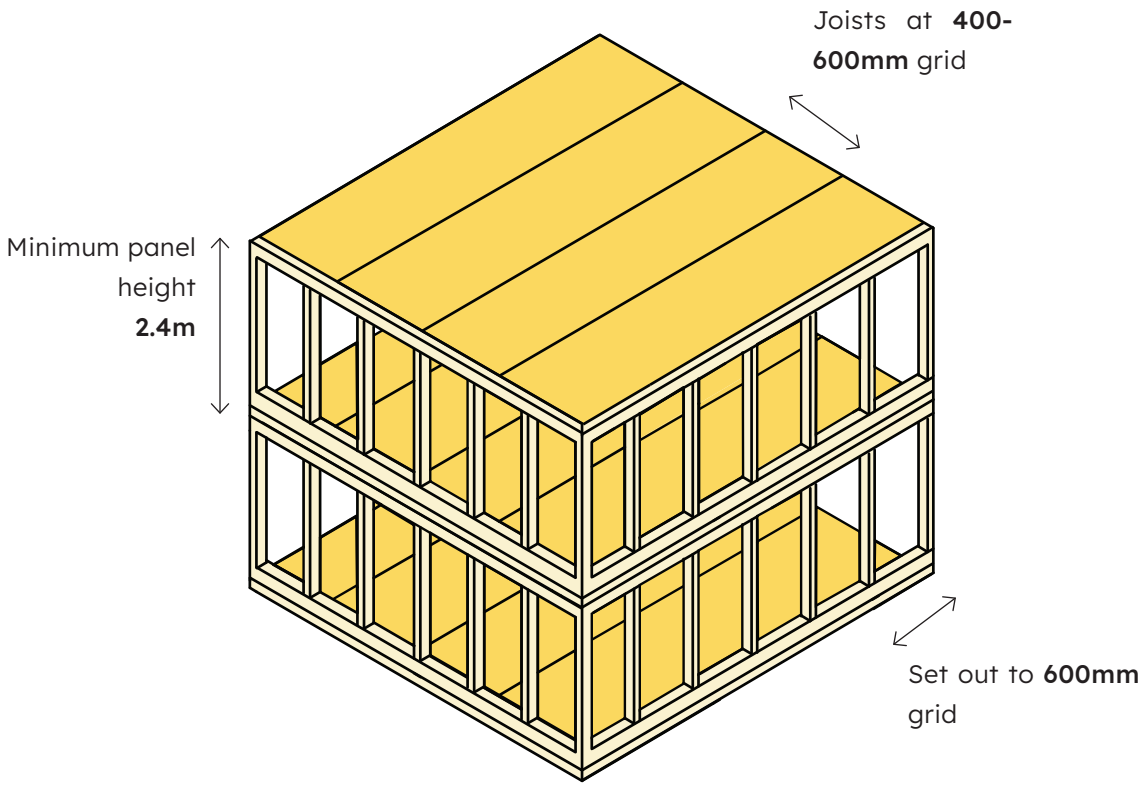
Wall, floor and roof panels are typically prefabricated: either as open panels (i.e. timber stud frame with sheathing typically applied to one face) or closed panels (i.e. timber stud frame pre-finished with insulation, linings, membranes and windows). Both approaches are classified as MMC Category 2 solutions, and can significantly expedite construction.

Balloon frame

In this method, similarly prefabricated panels, incorporating vertical timber studs, insulation and membranes, often span two or more floors. Intermediate floors are suspended from the face of the wall panels which become load-bearing.

Balloon frame panels require longer timber studs to construct and necessitates the installation of fire barriers at compartment lines. As a result, platform frame is the more common UK timber frame methodology.

Prefabricated lightweight panels are often combined with components like webbed joists, and mass timber beams (e.g. glulam beams, I-joists) to create hybrid solutions for open plan and non-residential applications.



Parameters of lightweight timber frame structural systems